IoT-Based Weed Detection & Automated Physical Cutting System

Project Pipeline, Team Roles and Cost Estimation (Major Project)

1. Team Roles (4 Members)

- Person 1 — Team Lead & AI Lead: project management, model selection, final report and demo supervision.
- Person 2 — Data Engineer / AI Implementation: dataset collection, labeling, training and inference scripts.
- Person 3 — Embedded / IoT Engineer: ESP32 firmware, cloud dashboard, wiring and communication.
- Person 4 — Mechanical / Hardware Engineer: cutter design, motor mounting, enclosure and system testing.

2. 8-Week Project Pipeline

Week	Tasks
1	Finalize scope, prepare BOM, order components.
2	Collect and label dataset; hardware arrival and bench tests.
3	Train baseline AI model; ESP32 communication testing.
4	Integrate detection with motor activation.
5	Add sensors and IoT dashboard.
6	Field-like testing and model refinement.
7	Long-run testing and documentation.
8	Presentation, viva prep and final demo.

3. Acceptance Criteria

- Detection precision ≥ 85%.
- Actuation success ≥ 90% when weed is within reachable region.
- IoT dashboard logs events with timestamps.
- System includes safety enclosure and emergency stop.

4. Cost Estimation — Tier A (Ultra-Low)

Item	Cost (■)
Webcam	600
ESP32	500
Relay Module	150
12V Motor	400
Cutting Blade	250
Mount & Bracket	200
12V Adapter	700
Wiring & Switches	500
Protective Shield	300

Sensors	200
Wooden Base	300
Misc	200
Subtotal	4300
Contingency	860
Total	5160

5. Cost Estimation — Tier B (Recommended)

Item	Cost (■)
Raspberry Pi 4	6500
Camera Module	1400
ESP32	500
Motor Driver	300
Relay	150
12V Motor	400
Cutting Blade	250
Battery	1200
DC-DC Converter	300
Wiring & Switches	500
Shield & Enclosure	400
Sensors	200
Chassis	800
Misc	500
Subtotal	13600
Contingency	2040
Total	15640

6. Risks & Mitigation

- False cuts: use high confidence threshold and manual override.
- Blade safety: protective enclosure and timed activation.
- Lighting issues: retrain model with augmented images.
- Power issues: fuse protection and current testing.

7. Milestones

- Dataset ≥ 500 labeled images.
- Detection accuracy ≥ 85%.
- End-to-end demo recorded.
- Wiring diagram and report completed.